LASSO for the force constants with exact invariances (FOCEX, `enforce_inv` = 2 and 3)

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Abstract

FOCEX fits the force constants (FCs) to force–displacement data while imposing the translational, rotational and Huang invariances. With `enforce_inv` = 2 and 3 the fit is a LASSO: an L_1 penalty on the FCs selects the terms the data support. This note gives the formulation, the numerical steps and the reason for each choice, and lists the tests the code has passed. In short: the penalty acts on the FCs themselves, the invariances hold exactly because $\mathbf{x}$ is confined to the kernel of the invariance matrix, that kernel is first cut to the directions the data resolve (the same cut the SVD fit makes), the LASSO is solved by an interior-point method, and, for `enforce_inv` = 3, the selected FCs are refitted by least squares. The L_1 weight is chosen by cross-validation.

1 The problem

The unknowns are the n_x irreducible FCs $\mathbf{x}$ of all ranks fitted (the columns of the fit). The data give N force equations and the invariances n_C homogeneous ones,

$$\mathbf{A}\mathbf{x} \approx \mathbf{b} \quad (N \times n_x), \qquad \mathbf{C}\mathbf{x} = 0 \quad (n_C \times n_x). \tag{1}$$

$\mathbf{C}$ is rank deficient and has many redundant rows (exact cubic test model: $n_C = 1882$, rank 379, $n_x = 555$). Its kernel has an orthonormal basis $\mathbf{K}$ ($n_x \times n_K$, from the SVD of $\mathbf{C}$, `get_kernel`), and every $\mathbf{x} = \mathbf{K}\mathbf{y}$ satisfies the invariances exactly.

`enforce_inv` = 1 (SVD, unchanged). $\min_{\mathbf{y}} \|\mathbf{A}\mathbf{K}\mathbf{y} - \mathbf{b}\|$ by `svd_set`: the SVD of $\mathbf{A}\mathbf{K}$, with singular values below `svdcut` $\sigma_{\max}$ discarded, i.e. the minimum-norm least-squares solution. `svdcut` = `force_error`$^2 = 10^{-10}$ by default.

2 What was wrong with the previous LASSO

The previous `enforce_inv` = 2 minimised $\frac{1}{2}\|\mathbf{A}\mathbf{K}\mathbf{y} - \mathbf{b}\|^2 + \mu \sum_k n_k |y_k|$: the penalty was on the kernel coefficients $\mathbf{y}$. The columns of $\mathbf{K}$ are dense orthonormal mixtures of many FCs, so FCs that are sparse are dense in $\mathbf{y}$, and a sparse $\mathbf{y}$ gives dense FCs. On an exact model (a pair potential, below) the FC2/FC3/FC4 had 98/80/376 non-zero terms where the true FCs have 11/20/43, and FC4 was wrong (on-site -0.43 for 288). Its solver (coordinate descent) also never converged.

Alternatives that were tested and rejected.

- *Elimination basis* (row-reduce $\mathbf{C}$, the free FCs are a subset of $\mathbf{x}$; ALAMODE). With the full constraint set the result depends on which FCs are eliminated (those carry no penalty) and FC4 came out wrong (error 28 of 288). ALAMODE, with LASSO, keeps only the

constraints inside each rank and drops the rotational rows coupling FC2–FC3 and FC3–FC4 (its own warning, `alm/constraint.cpp`); that left FC4 errors up to 1.2×10^4.

- *Free LASSO on* $\mathbf{x}$*, then a constrained least-squares fit on the kept FCs.* If the first step misses one true FC, the constraint rows containing it force its partners to zero in the second: the reduced kernel had dimension 3 and FC3 = FC4 = 0.

3 Formulation

$$\min_{\mathbf{x}} \ \frac{1}{2N}\|\mathbf{A}\mathbf{x}-\mathbf{b}\|^2 + \lambda \sum_j w_j|x_j| \qquad \text{subject to} \qquad \mathbf{C}\mathbf{x} = 0. \tag{2}$$

The penalty is on the FCs. w_j is the rms of column j of $\mathbf{A}$ (standardisation, as in scikit-learn, MATLAB and ALAMODE): FCs of different rank carry different powers of length, and $w_j|x_j|$ is what column j contributes to the forces, so the penalty does not depend on units. Columns the data do not see at all (Ge: four columns at 2×10^{-20}, three exactly zero) get $w_j = 10^{-8} \max w$, see §5.

4 Step 1: the part of the kernel the data resolve

Directions of the kernel along which the forces do not change cannot be fitted. The SVD fit sets them to zero (minimum norm). The LASSO must do the same: without it, FCs invisible in the data carry almost no penalty, and through the invariance couplings they absorb arbitrary values (Ge: phonons of $10^8\,\mathrm{cm}^{-1}$). So with

$$\mathbf{A}\mathbf{K} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \qquad \mathbf{K}_r = \mathbf{K}\mathbf{V}_r, \quad \mathbf{V}_r = \{\mathbf{v}_k : \sigma_k > \texttt{svdcut}\,\sigma_{\max}\}, \tag{3}$$

$\mathbf{x}$ is restricted to $\mathbf{x} = \mathbf{K}_r\mathbf{y}$. This is the cut `svd_set` makes, and $\mathbf{A}\mathbf{K}_r$ has full column rank. Exact cubic model: 159 of 176 directions kept; the 17 removed are FC2 components on the boundary of the supercell's Wigner–Seitz cell that cancel in every supercell force. Ge: 134 of 144.

Why not $\mathbf{A}^T\mathbf{A}$. $\mathbf{K}^T\mathbf{A}^T\mathbf{A}\mathbf{K}$ is small ($n_K \times n_K$) and quick to form, but its eigenvalues are σ_k^2: the cut $\sigma_k/\sigma_{\max} = 10^{-10}$ becomes 10^{-20}, below the double-precision resolution 2×10^{-16}. From $\mathbf{A}^T\mathbf{A}$ only $\sigma_k/\sigma_{\max} \gtrsim 10^{-8}$ can be resolved. Instead $\mathbf{A}\mathbf{K}$ is reduced row by row to a triangular $\mathbf{R}$ ($n_K \times n_K$) by Givens rotations, $\mathbf{A}\mathbf{K} = \mathbf{Q}_0\mathbf{R}$; $\mathbf{R}$ has the same singular values and right singular vectors as $\mathbf{A}\mathbf{K}$, and its SVD is cheap. Cost: about Nn_K^2 operations (the same as forming $\mathbf{A}^T\mathbf{A}$), memory n_K^2; $\mathbf{A}\mathbf{K}$ is never stored.

5 Step 2: variables in which the problem is simple

Let $\mathbf{W} = \mathrm{diag}(w_j)$ and $\mathbf{z} = \mathbf{W}\mathbf{x}$ (standardised FCs). Take an orthonormal basis $\mathbf{Q}$ of the range of $\mathbf{W}\mathbf{K}_r$ (its left singular vectors, $n_x \times r$, $\mathbf{Q}^T\mathbf{Q} = \mathbf{1}$), so $\mathbf{z} = \mathbf{Q}\mathbf{t}$ and $\mathbf{x} = \mathbf{W}^{-1}\mathbf{Q}\mathbf{t}$. Then (2) becomes, with no constraint left,

$$\min_{\mathbf{t}} \ \tfrac{1}{2}\mathbf{t}^T\mathbf{G}\mathbf{t} - \mathbf{c}^T\mathbf{t} + \lambda\|\mathbf{Q}\mathbf{t}\|_1, \qquad \mathbf{G} = \tilde{\mathbf{A}}^T\tilde{\mathbf{A}}, \ \ \mathbf{c} = \tilde{\mathbf{A}}^T\mathbf{b}/\sqrt{N}, \ \ \tilde{\mathbf{A}} = \mathbf{A}\mathbf{W}^{-1}\mathbf{Q}/\sqrt{N}. \tag{4}$$

$\mathbf{G}$ is $r \times r$ and is accumulated row by row, once per cross-validation fold (§8). The L_1 term is on $\mathbf{z}$, i.e. on the FCs; $\mathbf{Q}$ being orthonormal keeps the Newton systems below well scaled. Above $\lambda_{\max} = \|\mathbf{Q}\mathbf{c}\|_\infty$ the solution is $\mathbf{t} = 0$ exactly: $\boldsymbol{\nu} = \mathbf{Q}\mathbf{c}$ satisfies the optimality condition $\mathbf{Q}^T\boldsymbol{\nu} = \mathbf{c}$ with $|\nu_i| \le \lambda$.

Weight floor. $\mathbf{x} = \mathbf{W}^{-1}\mathbf{Q}\mathbf{t}$ divides by w_j. For a column at 10^{-20} the round-off of $\mathbf{Q}$ (10^{-16}) became FC errors of order 10^4 (Ge: an FC2 of 153 where the SVD has 10^{-6}). Hence $w_j \geq 10^{-8} \max w$; genuine Ge columns go down to $3 \times 10^{-6} \max w$, so only the round-off zeros are affected.

6 Step 3: solving the LASSO

(4) is the quadratic program

$$\min \ \tfrac{1}{2}\mathbf{t}^T\mathbf{G}\mathbf{t} - \mathbf{c}^T\mathbf{t} + \lambda \mathbf{1}^T\mathbf{s}, \qquad \mathbf{a} = \mathbf{s} - \mathbf{Q}\mathbf{t} \geq 0, \quad \mathbf{b}' = \mathbf{s} + \mathbf{Q}\mathbf{t} \geq 0, \tag{5}$$

solved by a primal–dual interior-point method (Mehrotra predictor–corrector). With multipliers $\boldsymbol{\alpha}, \boldsymbol{\beta} \geq 0$ the optimality conditions are $\mathbf{G}\mathbf{t} - \mathbf{c} + \mathbf{Q}^T(\boldsymbol{\alpha} - \boldsymbol{\beta}) = 0$, $\boldsymbol{\alpha} + \boldsymbol{\beta} = \lambda$, $\alpha_i a_i = \beta_i b'_i = 0$. Eliminating $\mathbf{s}, \boldsymbol{\alpha}, \boldsymbol{\beta}$ from the Newton step leaves one $r \times r$ system,

$$(\mathbf{G} + \mathbf{Q}^T\mathbf{D}\mathbf{Q})\,\delta\mathbf{t} = \text{r.h.s.}, \qquad D_i = \frac{4A_iB_i}{A_i + B_i}, \ A_i = \alpha_i/a_i, \ B_i = \beta_i/b'_i, \tag{6}$$

which is positive definite ($\mathbf{D} > 0$, $\mathbf{Q}$ full column rank) even where $\mathbf{G}$ is badly conditioned; it is solved by Cholesky. The iteration stops when the residuals of the two equations and the duality gap $\boldsymbol{\alpha}^T\mathbf{a} + \boldsymbol{\beta}^T\mathbf{b}'$ are below 10^{-10} (relative). It takes 10–16 iterations.

Why this solver. ADMM (the standard splitting for $\|\mathbf{Q}\mathbf{t}\|_1$) had not converged after 200 000 iterations at 3 of 5 values of λ on the exact model. Proximal-gradient methods (ISTA, FISTA) need the proximal map of $\lambda\|\mathbf{Q}\mathbf{t}\|_1$, which has no closed form; on the unconstrained FOCEX matrix FISTA stopped with optimality violations of 0.3–7 % at its default settings. LARS is exact and fastest for the unconstrained LASSO, but does not handle the constraint. The interior point gives the objective to 10^{-11} of an independent conic solver (cvxpy/Clarabel) at every λ.

The selected FCs. $S = \{j : |z_j| > 10^{-8} \max |z|\}$. Interior-point zeros are small, not exact; at small λ a few terms that vanish in the exact solution pass the test (against LARS, unconstrained: 57 kept for 46 at $\lambda = 10^{-4}\lambda_{\max}$; no true term lost). The test errs on the side of keeping a term, deliberately: a lost true term makes the refit collapse (§2), an extra one is fitted to about zero by it.

7 Step 4 (`enforce_inv` = 3): least-squares refit on the selected FCs

The LASSO shrinks every kept FC towards zero by an amount proportional to λ. The refit removes this bias: least squares over the FCs in S, all others zero, invariances exact,

$$\min_{\mathbf{t}} \ \tfrac{1}{2}\mathbf{t}^T\mathbf{G}\mathbf{t} - \mathbf{c}^T\mathbf{t} \quad \text{with} \quad (\mathbf{Q}\mathbf{t})_j = 0, \ j \notin S. \tag{7}$$

The admissible $\mathbf{t}$ form the null space $\mathbf{N}$ of $\mathbf{Q}_{S^c,:}$, computed by a Householder QR of $\mathbf{Q}^T_{S^c,:}$ with column pivoting (an SVD routine did not converge on some of these matrices, whose singular values cluster at 0 and 1). Then $\mathbf{t} = \mathbf{N}\mathbf{s}$, with $\mathbf{s}$ the pseudo-inverse solution of $(\mathbf{N}^T\mathbf{G}\mathbf{N})\mathbf{s} = \mathbf{N}^T\mathbf{c}$, with eigenvalues below 10^{-13} of the largest dropped. *Limitation:* this step works with the Gram matrix, so its own cut corresponds to $\sigma/\sigma_{\max} \approx 3 \times 10^{-7}$, not 10^{-10}; directions between the two are treated as unresolved here. On the tests below it agrees with a least-squares solution on $\mathbf{A}$ itself (numpy) to 4×10^{-9}.

When to refit. With accurate forces the refit gives the unbiased FCs (exact on the exact model). With noisy forces or too few snapshots it fits the selected terms to the noise; then `enforce_inv` = 2 (no refit) is the safer choice.

8 Step 5: choosing λ by cross-validation

The force rows are split into 5 folds, interleaved (row i in fold $1 + \mathrm{mod}(i-1, 5)$), so that every fold samples every snapshot. For each fold the training Gram matrix is the total minus the fold's, so the data are read once. On each training set the LASSO is solved on a grid of 40 values of λ, log-spaced from $\lambda_{\max}$ to $10^{-4}\lambda_{\max}$ (from the previous solution as starting point), and the mean squared error on the held-out rows is recorded for the model that will be returned (refitted or not). The λ with the smallest mean error is chosen; within $10^{-10}\langle b^2\rangle$ of the minimum (round-off, exact data) the largest such λ is taken. The final FCs are the fit on all rows at that λ. If the minimum is at the end of the grid, a warning is printed.

9 Tests

Exact model. Pair potential $A(r_c^2 - r^2)^2$, fcc lattice with two species, ranks 2–4 fitted; its FCs are known exactly (FC2/FC3/FC4: 11/20/43 non-zero of 99/80/376). Forces from FOCEX's own matrix.

	FC2 max error	FC3 max error	FC4 max error	non-zero FC2/3/4
penalty on **y** (LARS solver)	28.8	0.67	288	98/80/376
`enforce_inv` = 3, exact forces	9×10^{-10}	4×10^{-9}	8×10^{-7}	11/20/43
1 % random noise on the forces:				
`enforce_inv` = 1 (SVD)	0.095	1.13	206	98/80/376
`enforce_inv` = 2	0.012	1.35	9.2	54/61/350
`enforce_inv` = 3	0.017	0.58	4.8	11/32/136

(largest FCs: FC2 80, FC3 48, FC4 288). With noise this data set does not determine FC4 well for any method; the LASSO reduces the error by a factor 20–40 compared with the SVD.

Ge (FOCEX example, DFT forces). SVD: fit 0.036 %, 171 of 173 FCs non-zero. `enforce_inv` = 3: fit 0.044 %, 107 non-zero; FC2 within 2.5×10^{-4} and FC3 within 0.12 (largest 28) of the SVD values.

Solver checks. Interior point vs. cvxpy/Clarabel at five values of λ: objective equal to 10^{-11}; refit vs. numpy least squares: 4×10^{-9}.

10 Open points

- The refit's Gram-matrix cut (§7) is coarser than `svdcut`; a QR-based refit would remove the difference.
- The selection threshold 10^{-8} (§6) is a numerical choice, tested only on the cases above.
- BZO and MoTe2 with the corrected code are still to be run, with the phonon dispersions.

References. B. Efron, T. Hastie, I. Johnstone, R. Tibshirani, Ann. Statist. **32**, 407 (2004) (LARS); S. Boyd *et al.*, Found. Trends Mach. Learn. **3**, 1 (2011) (ADMM); N. Meinshausen, Comput. Stat. Data Anal. **52**, 374 (2007) (relaxed lasso: refit after selection; cited from memory, not checked); S. Mehrotra, SIAM J. Optim. **2**, 575 (1992) (predictor–corrector interior point; cited from memory, not checked); T. Tadano, ALAMODE, `github.com/ttadano/alamode` (constraint handling).